

Maximal minors scaled by the reference ball volume

MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors

def

Fixes the volume factor and increasing-row signs.

Statement

Let $n, N \in \mathbb{N}_0$ and give $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ their coordinate sup norms. Let λ_n be the fixed coordinate Lebesgue measure on E . Let $p : E \rightarrow \mathbb{R}$ be a continuous real seminorm with specified constants $a_p, b_p > 0$ such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in E).$$

The lower bound makes p a norm. Its closed unit ball and real reference volume are

$$K_p = \{x : p(x) \leq 1\}, \quad v_p = \lambda_n(K_p) \in (0, \infty).$$

Here and below a finite Lebesgue measure is read as a real number when used as a scalar.

Let $A : E \rightarrow F$ be linear. Index its maximal minors by increasing row selections $I = (i_1 < \dots < i_n)$ from $\{1, \dots, N\}$, using $A_{ij} = (Ae_j)_i$. Write $\Pi_n(A)$ for the vector of these maximal minors and $\mathcal{V}_p(A)$ for its volume-scaled version. They are given by

$$\mathcal{V}_p(A) = v_p \Pi_n(A), \quad (\mathcal{V}_p(A))_I = v_p \det(A_{i_a, j})_{a, j=1}^n.$$

Definition

This definition multiplies every maximal minor by v_p . Increasing row order fixes each determinant sign. The real ball-volume definition takes the finite measure $\lambda_n(K_p)$ and reads it as the real scalar v_p . The volume is finite and positive because

$$\{x : \|x\|_\infty < 1/b_p\} \subseteq K_p \subseteq \{x : \|x\|_\infty \leq 1/a_p\}.$$

The first inclusion puts zero in the interior (interior-point lemma); continuity makes the ball closed and the second bounds it, giving compactness. Positivity of Lebesgue measure on a nonempty open set and finiteness on compact sets give finite positive ball volume.

Assumptions

The source and target are finite real coordinate spaces with sup norms. The seminorm p is continuous and has the stated positive lower and upper comparison constants. The measure is the original coordinate Lebesgue measure. No contraction assumption is imposed on A . The dimensions n, N may be zero. When $n = 0$, the

unique empty minor has determinant 1; when $N < n$, the minor-index set is empty and its function space has one element. Neither case is excluded.

Conclusion

The output is a real vector with one coordinate for each increasing row selection I . The reference coordinates and measure stay fixed.

Every linear map between these finite-dimensional spaces is continuous. The type of the source therefore does not narrow the stated class of maps.

Main citations

- Exact declaration and its source — `MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors`
- The real ball-volume definition — `MathlibAnnex.EquivalentSeminorm.closedUnitBallVolume`
- interior-point lemma — `MathlibAnnex.EquivalentSeminorm.zero_mem_interior_closedUnitBall`
- compactness — `MathlibAnnex.EquivalentSeminorm.isCompact_closedUnitBall`
- finite positive ball volume — `MathlibAnnex.EquivalentSeminorm.closedUnitBallVolume_pos`

Lean source signature (exact)

```
-- The maximal-minor vector multiplied by the model ball's Lebesgue volume. -/
def ballVolumeScaledMaximalMinors {n N : ℕ}
  (M : EquivalentSeminorm (Fin n → ℝ))
  (A : (Fin n → ℝ) →L[ℝ] (Fin N → ℝ)) : MaximalMinorIndex n (Fin N) → ℝ :=
  M.closedUnitBallVolume • maximalMinors (LinearMap.toMatrix' A.toLinearMap)
```

Read the parameters and the defining expression as follows.

In the source	Mathematical meaning
<code>{n N : ℕ}</code>	Arbitrary nonnegative integers n, N . Braces mean that Lean may infer these parameters; they are not extra assumptions.
<code>Fin n → ℝ; Fin N → ℝ</code>	The spaces $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ with their sup norms. A vector is a list of real coordinates, indexed from 0 in Lean and from 1 in the formulas here.
<code>M : EquivalentSeminorm (Fin n → ℝ)</code>	The continuous seminorm $p = M.p$, with $a_p = M.lower > 0$, $b_p = M.upper > 0$ and $a_p \ x\ _\infty \leq p(x) \leq b_p \ x\ _\infty$ for every $x \in E$.
<code>A : ... →L[ℝ] ...</code>	A continuous real-linear map $A : E \rightarrow F$. The notation $\rightarrow L[\mathbb{R}]$ asserts linearity and continuity; it does not assert a contraction bound.
<code>MaximalMinorIndex n (Fin N) → ℝ</code>	A real vector indexed by the increasing row selections $I = (i_1 < \dots < i_n)$ from $\{1, \dots, N\}$.

In the source	Mathematical meaning
<code>M.closedUnitBall; M.closedUnitBallVolume</code>	$K_p = \{x : p(x) \leq 1\}$ and $v_p = \lambda_n(K_p) > 0$, with the finite measure read as a real number.
<code>A.toLinearMap</code>	The same map A , retaining its linearity and forgetting only the stored continuity proof.
<code>LinearMap.toMatrix' A.toLinearMap</code>	The $N \times n$ matrix $A_{ij} = (Ae_j)_i$ in the standard coordinate bases.

In the source	Mathematical meaning
<code>maximalMinors (...)</code>	The vector $\Pi_n(A)$ with coordinate $\det(A_{i_a,j})_{a,j=1}^n$.
<code>M.closedUnitBallVolume • ...</code>	Multiply every coordinate of that vector by v_p . Thus the output is $\mathcal{V}_p(A)$; $:=$ introduces this defining equation.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 12–14:

```

noncomputable section

namespace MathlibAnnex.Matrix

```

▼ Exact content identity

Declaration: `MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors`
Accepted content SHA-256: 6fac7ef9b6a4689ca91d0f1aadb7686741c672e74c08bdd9054d7ee09f56289c
Accepted source guide SHA-256: 07eee092d07645f1f4c408179c68c30125447309b9e7bf0bb589221a51f365f8
Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73